



To: Dr. Ferguson and the Editorial Board of *JEP:LMC*

We are resubmitting our manuscript (XLM-2025-0564), entitled “Compensation in audiovisual speech perception: discounting the pen in the mouth.” We are grateful to you and the reviewers for your highly constructive feedback on our initial submission. We were heartened to see that the reviewers “found the study to be interesting, well-executed and potentially impactful for understanding audiovisual speech perception” and, in particular, that reviewers considered the focus on non-linguistic visual information influencing speech perception clearly motivated and valuable.

We very much appreciate the feedback from reviewers, which helped us address the presentational points they raised. Specifically:

1. The revised introduction now clarifies our contributions to previous work that has used similar audiovisual manipulations as part of experiments on lexically-guided perceptual recalibration (especially, Kraljic et al. 2008, as well as Kraljic & Samuel, 2011; Liu & Jaeger, 2018).

In short, these studies investigated how a visually-presented pen in the mouth (vs. hand) of a speaker during the lexically-guided exposure phase affected listeners perception during a subsequent *audio-only* test phase. The thought-provoking finding of these studies was that perceptual recalibration can be blocked or reduced if the audio-visual exposure items with shifted pronunciations occur only while the speaker has a pen in the mouth. Competing explanations have been offered for this reduction of perceptual recalibration—from causal inferences (e.g., Kraljic et al., 2008; Liu & Jaeger, 2018) to separate exemplar storage of audio-only vs. audio-visual inputs (Kraljic & Samuel, 2011).

Critically, the design of these studies did not allow researchers to estimate how the placement of the pen affects perception *in that moment*: the pen was only ever displayed during the *exposure* phase, which used a lexical decision—rather than categorization—task. This makes it difficult to adjudicate between the competing hypotheses about the pen’s downstream effects during the test phase. The revised introduction now clarified that this is the knowledge gap addressed by our study, which is the first study to use audiovisual test continua that manipulate the location of the pen (mouth vs. hand). We have also revised the discussion to spell out in more detail what our results mean for this earlier line of work.

2. The revised methods now contain all the information requested by reviewers (moved there from the supplementary information). We agree that this makes the manuscript more accessible. The reviewers’ comments also made us realize that we had made a small mistake in reporting the exclusion criteria. This is now fixed, and it didn’t change any of the results.

3. The revised discussion of Study 2 now transparently acknowledges the concern raised by Reviewer 1. In that context, we now also elaborate on participant responses to the post-experiment survey, and on Study 2b (described in detail in the SI). These both ameliorate—but do not fully remove—that concern. This revision, we hope, helps readers to maintain an adequate level of uncertainty about our interpretation of our results.

We have also fixed the typos pointed out by all reviewers; we thank them for their attention and have given the manuscript another careful read ourselves to eliminate further mistakes.

The response to reviewers document included in this resubmission includes a reproduction of the summary of changes above, as well as a point-by-point response to the reviewers' comments.

Sincerely,

Shawn Cummings, Gevher Karboga, Menghan Yang, and Florian Jaeger